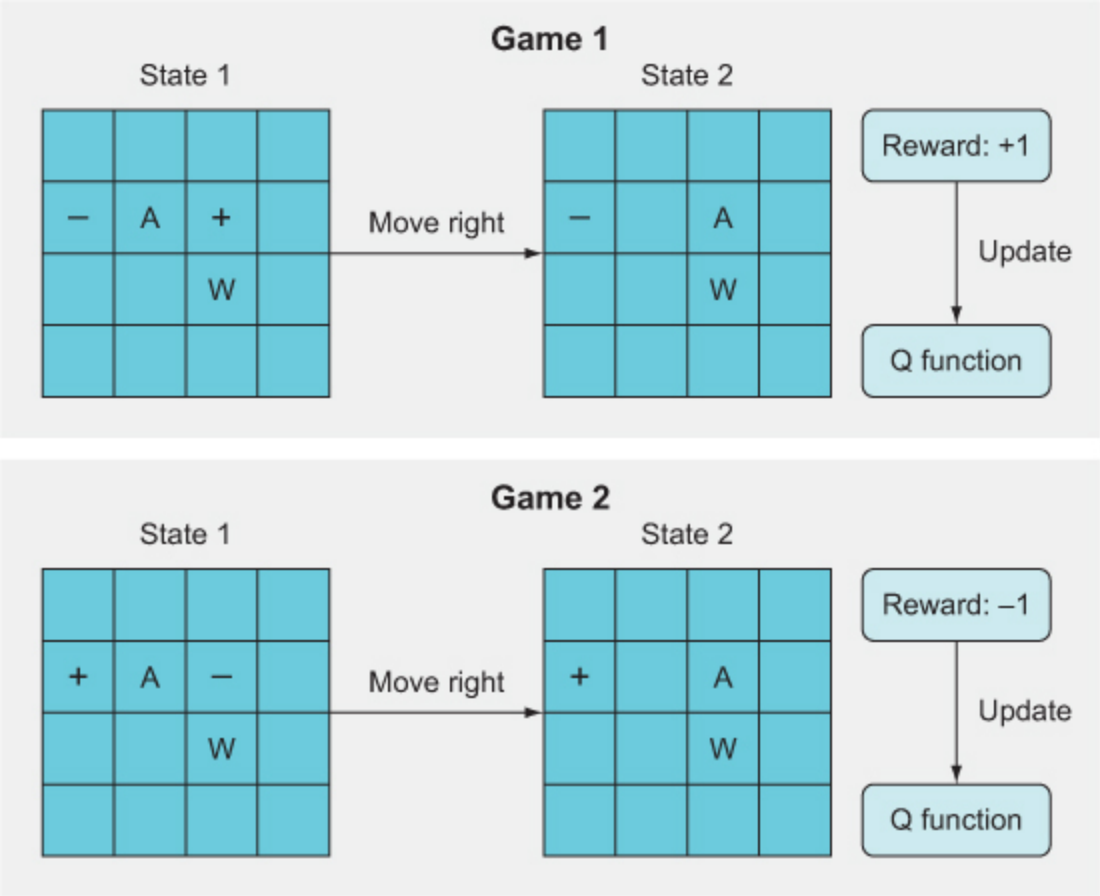


Now game 2 is initialized and the player is again between the goal and pit, but this time the goal is on the *left* and the pit is on the right. Perhaps to our naive algorithm, the state *seems* very similar to the last game. Since last time moving right gave a nice positive reward, the player chooses to make one step to the right again, but this time it ends up in the pit and gets –1 reward. The player is thinking, "What is going on? I thought going to the right was the best decision based on my previous experience." It may do backpropagation again to update its state-action value, but because this state-action is very similar to the last learned state-action, it may override its previously learned weights.

This is the essence of catastrophic forgetting. There's a push-pull between very similar state-actions (but with divergent targets) that results in this inability to properly learn anything. We generally don't have this problem in the supervised learning realm, because we do randomized batch learning where we don't update our weights until we've iterated through some